A logarithmic-budget sieve for Erdős problem #647

An explicit sparsity bound with a Lean formalization

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Abstract

Erdős and Selfridge asked whether any integer $n > 24$ satisfies $\tau(n-k) \le k+2$ for every $1 \le k < n$, where τ counts positive divisors. Let $C(X)$ count integers satisfying this full condition up to X. We give a proof, accompanied by a Lean formalization, that for every fixed $0 < c < 1499/10^6$ and all sufficiently large natural X,

$$C(X) \le X \exp\left(-c\,\frac{(\log X)^{\log 2/(1+\log 2)}}{\log\log X}\right).$$

The argument converts the divisor restrictions into a logarithmic budget for distinct prime factors, subtracts unavoidable small-prime occurrences, and uses a finite weighted Bonferroni estimate with uniform Chinese-remainder counting errors. Mertens convergence, factorial bounds, and explicit parameter asymptotics then yield a positive mass–budget gap. All amplified error terms are retained through the final absorption. The coefficient $1/1000$ is an immediate corollary, not an optimization target. We describe the ChatGPT–operator workflow, the formalization's trust boundary, and reusable components, and identify possible connections to other Erdős problems without claiming new results for them. The supplied final verification run passes all 30 configured gates. The theorem is a sparsity result: it proves neither finiteness nor the nonexistence of candidates beyond 24.

Keywords. Divisor function; consecutive integers; upper-bound sieve; Bonferroni inequalities; Chinese remainder theorem; Mertens theorem; Lean.

Contents

1 The problem and its arithmetic form

For a positive integer m, let $\tau(m)$ be the number of positive divisors of m, let $\omega(m)$ count its distinct prime divisors, and let $\Omega(m)$ count its prime factors with multiplicity. All logarithms in this paper are natural. Erdős's 1979 discussions of unconventional problems in number theory include the question now catalogued as problem #647, posed with Selfridge [3, 7, 8]:

> Is there an integer $n > 24$ such that $\max_{1\le m<n}\bigl(m+\tau(m)\bigr) \le n+2$?

The value $n = 24$ satisfies the inequality; the question asks for an example beyond it. Substituting $m = n - k$ makes the same condition a collection of bounds on a backward window of consecutive integers.

Definition 1.1. A *candidate* is a natural number $n > 24$ such that

$$\tau(n-k) \le k+2 \qquad (1 \le k < n). \tag{1.1}$$

Write C for this set, and for $X \in \mathbb{N}$ put $C(X) = \#(C \cap [1, X])$.

Here "candidate" means an integer satisfying the *entire* condition, not merely one surviving a finite computation. The first restrictions are $\tau(n-1) \le 3$, $\tau(n-2) \le 4$, and $\tau(n-3) \le 5$; every further shift is also constrained. For the proof we may retain only the first H restrictions, provided $n > H$. This loses information, but gives an upper bound for the actual candidate count.

1.1 Position relative to existing work

Finite exclusion and asymptotic counting address different parts of the problem. Mian and Siddique give a Lean exclusion certificate for $24 < n \le 10^9$, with an audited axiom closure and additional replay checks reported in their work [15]. Our argument is not a larger exclusion computation and does not subsume that finite assertion: its onset is not numerically specified.

Hughes's consolidation manuscript states both a seven-form bound $C(X) \ll X/(\log X)^7$ and a growing-window bound

$$C(X) \le X\exp\bigl(-(\log\log X)^{2-o(1)}\bigr), \tag{1.2}$$

with proofs of those density statements attributed there to companion manuscripts [12]. The saving in the present theorem is asymptotically larger than the saving displayed in (1.2). Our arithmetic starting point is the logarithmic bound $\omega(m) \le \log\tau(m)/\log 2$, together with an explicit debit for small primes. This differs from the quadratic-in-H budget obtained by summing the weaker implication $\Omega(m) + 1 \le \tau(m)$ over the window. The comparison is with the stated result in that inspected manuscript; it is not a claim of an exhaustive priority determination.

Broader work on prime factors of consecutive integers includes the probabilistic and sieve arguments of Tao–Teräväinen and Lau [14, 18]. We return to specific related problems in Section 13. The classical background is upper-bound sieve theory and weighted inclusion–exclusion [9, 10]. The components used here are classical in origin. The contribution described by this paper is their constant-sensitive assembly for (1.1), the resulting bound, and a checked implementation with explicit dependency boundaries. We do not assert optimality of the exponent or of the numerical coefficient.

2 The proved result and its scope

Set

$$a = \frac{\log 2}{1+\log 2}, \qquad \kappa = \frac{1499}{10^6}, \qquad S(X) = \frac{(\log X)^a}{\log\log X} \quad (X \ge 3). \tag{2.1}$$

Thus $0 < a < 1$ and $a \approx 0.4094$.

Theorem 2.1 (Fixed-coefficient sparsity)**.** *For every fixed real c satisfying $0 < c < \kappa$, there is a natural number $X_0 = X_0(c) \geq 3$ such that, for every natural $X \geq X_0$,*

$$\boxed{C(X) \leq X \exp\left(-c\,\frac{(\log X)^a}{\log\log X}\right).} \tag{2.2}$$

In particular, (2.2) *holds with $c = 1/1000$.*

The main Lean declaration is `Erdos647Sieve.Endpoint.endpointBound_of_lt_principal_rate`. It actually permits any real $c < \kappa$; the positive range is the useful sparsity statement. The explicit corollary is `Erdos647Sieve.Endpoint.endpointBound_one_div_thousand`; the original fixed-coefficient specification is discharged by `Erdos647Sieve.endpoint` [11].

The quantifiers are essential: c is selected *before* the onset X_0 and the variable X. The theorem does not allow a coefficient $c(X)$ tending to zero, does not provide a numerical X_0, and does not give one common onset for all c approaching κ. Its coefficient range is open at κ. Section 11 explains why the present majorant needs strict slack.

Corollary 2.2 (Consequences, not an existence decision)**.** *The candidate set has natural density zero. For every fixed $M > 0$, $C(X) \leq X/(\log X)^M$ for all sufficiently large natural X. Nevertheless, the right-hand side of* (2.2) *tends to infinity.*

Proof. Put $Q = \log X$ and $L = \log Q$. Then $S = Q^a/L \to \infty$, and $S/L = Q^a/L^2 \to \infty$. Consequently $e^{-cS} \leq e^{-ML} = Q^{-M}$ eventually. On the other hand, $S/Q = Q^{a-1}/\log Q \to 0$, so $\log(Xe^{-cS}) = Q - cS \to \infty$. □

These elementary consequences explain the scope of the theorem; they are not advertised as additional exported declarations in the recorded release. Even an infinite sufficiently sparse set is compatible with the bound. In particular, neither the existence nor the nonexistence of one candidate beyond 24 follows.

The proof has three levels. A *finite*, translation-uniform moment bound uses only elementary arithmetic and finite combinatorics. An *analytic* layer evaluates the available prime mass as the parameters grow. An *absorption* layer keeps all error terms and converts the resulting multi-term inequality into (2.2). This separation is reflected in the Lean packages, rather than being merely expository.

3 Methodology: AI-assisted construction and formal checking

3.1 Author, operator, and workflow

The development was conducted through ChatGPT at `https://chatgpt.com` [16]. The operator reports using the interface label "GPT-6 Pro", referred to in this project as Astra. We use the requested author attribution *GPT 6 Astra*, with *Justin V. Wick* designated as *operator*. This is an independent project, with no institutional affiliation. The interface label is workflow metadata supplied by the operator, not a reproducibly pinned model binary.

The conversational assistant proposed arguments, inspected library interfaces, produced Lean source and packaging changes, and revised them in response to compiler output. The operator selected the research direction, ran the project on a local machine, returned complete diagnostic archives, and managed the repository and pull-request boundaries. The workflow was iterative rather than an autonomous, single-call proof search. We make no controlled experimental claim about model success rates, compute efficiency, or reproducibility of the same conversation from a prompt.

The mathematical assurance is not the language model's confidence. Proposed identities and inequalities were expressed as Lean declarations, compiled with pinned dependencies, and checked against explicit statement and axiom targets. The ordinary-language proof below is a reconstruction of those ingredients; it should be read together with the theorem-to-source correspondence in Appendix B.1. The recorded artifacts, rather than an attempt to replay the model's internal generation, are the reproducibility object.

3.2 Reuse before reconstruction

The formalization uses Lean 4 and Mathlib [6, 19, 20]. Existing results on factorials, finite Euler products, harmonic sums, floor and ceiling limits, real powers, and integration were reused where their exact hypotheses fit. Project-specific adapters handle such issues as real casts of factorials, integer division versus real division, and equality between a named function and the expression appearing beneath a derivative.

The mathematical specification was kept separate from implementation repairs. For example, B_H is signed throughout, repeated weight values are not identified as indices, and translated intervals may begin at a negative integer. Those are statement-level choices. Changing a tactic or a module path does not license changing any of them.

A toolchain migration also exposed a practical limit of indiscriminate reuse: Lean, Mathlib, and upstream projects must agree on their dependency versions. The final artifact uses PNT+'s compatible Lean/Mathlib 4.32.2 environment. The finite package has only Mathlib dependencies. The analytic package isolates its PNT+ adaptation behind one compatibility module. Thus a consumer of the finite theorem does not inherit the analytic dependency.

3.3 What the verification evidence establishes

The supplied v48 record [11] reports 30 successful configured gates, 235 distinct declarations with transitive axiom reports, and 220 exact-type or definition examples in 32 audit files. All audited axiom sets are contained in

$$\{\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}\}. \tag{3.1}$$

The successful run has no build or audit warnings, and the recorded source snapshots are stable. Appendix B gives the exact artifact and version identifiers.

These checks have distinct jobs. Compilation constructs and checks proof terms; expanded-type examples connect the terms to the intended mathematical statements; axiom inspection detects an admitted or otherwise unauthorized premise in their transitive dependencies. A theorem of the form "assuming the moment bound, the conclusion follows" would not count as a proof of the moment bound itself. Ordinary runner tests instead check engineering behavior and do not establish universal mathematical assertions.

The PNT+ repository contains unfinished material outside the slice used here. The retained 14-module PNT+ import subtree was separately scanned for admitted source, and the consumed PNT and integrability declarations were axiom-audited. The unfinished `ZetaSummary` catalogue is not imported. This is a claim about the retained dependency, not a declaration that the whole upstream repository is admission-free [13].

The evidence is an operator-executed build, exact-type, and transitive-axiom audit of the supplied snapshot. An independently implemented kernel replay and external mathematical peer review are separate validation levels and are not claimed for this project. This qualification applies once to the artifact as a whole; it is not an additional hypothesis in any theorem below.

4 From divisor restrictions to a signed logarithmic budget

Fix a natural $H \geq 1$ and a real $y > H$. Define

$$\mathcal{P}(H, y) = \{p \text{ prime} : H < p \leq y\}, \tag{4.1}$$

$$F_H(n) = \prod_{k=1}^{H}(n-k), \qquad T_{H,y}(n) = \#\{p \in \mathcal{P}(H,y) : p \mid F_H(n)\}, \tag{4.2}$$

$$Q_H = \sum_{k=1}^{H}\left\lfloor \frac{\log(k+2)}{\log 2} \right\rfloor, \qquad D_H = \sum_{\substack{p \leq H \\ p \text{ prime}}} \left\lfloor \frac{H}{p} \right\rfloor, \qquad B_H = Q_H - D_H. \tag{4.3}$$

We often write $T(n)$ when H, y are fixed. The subtraction defining B_H is in $\mathbb{Z}$, not truncated subtraction in $\mathbb{N}$.

Lemma 4.1 (The occurrence budget)**.** *If $n > H$ and $\tau(n-k) \leq k+2$ for $1 \leq k \leq H$, then*

$$T_{H,y}(n) \leq B_H. \tag{4.4}$$

More generally, for every integer $n > H$,

$$D_H + T_{H,y}(n) \leq \sum_{k=1}^{H} \omega(n-k). \tag{4.5}$$

Proof. Every subset of the distinct prime divisors of a positive integer m gives a distinct divisor, so $2^{\omega(m)} \leq \tau(m)$. Taking logarithms and using the integrality of $\omega(m)$ gives

$$\omega(n-k) \leq \left\lfloor \frac{\log(k+2)}{\log 2} \right\rfloor.$$

Thus the total number of prime–shift incidences is at most Q_H.

For a prime $p \leq H$, every H consecutive integers contain at least $\lfloor H/p \rfloor$ multiples of p. These mandatory incidences contribute at least D_H. Each selected prime counted by $T(n)$ divides at least one factor of $F_H(n)$ and contributes a further incidence. The small-prime and selected-prime sets are disjoint. Counting all these incidences proves (4.5); combining it with the preceding upper bound proves (4.4). □

Since $T(n) \geq 0$, a negative value of B_H rules out every such prefix survivor above H. In particular,

$$B_H < 0 \implies C(X) \leq H. \tag{4.6}$$

This simple branch must remain available. Proving eventual nonnegativity of the corrected budget is unnecessary for the endpoint.

The key improvement is the size of the available budget. Without the debit, Q_H is of order $H \log H$. Small primes consume order $H \log\log H$ occurrences regardless of where the window lies. Subtracting that forced cost is precisely what permits constant-level cancellation at the critical exponent chosen later. The proof is counting prime occurrences, not assuming independent factorization of neighboring integers.

5 A finite weighted moment estimate

The following theorem is the finite core. It is independent of the prime number theorem, Mertens convergence, and the later parameter choices.

Theorem 5.1 (Uniform translated finite moment)**.** *Let $A \in \mathbb{Z}$, let $X, H \geq 1$ be natural numbers, let $y > H$, let $0 < z < 1$, and let $J \geq 0$ be an even natural number. Put*

$$t = 1 - z, \qquad \lambda = H \sum_{p \in \mathcal{P}(H,y)} \frac{1}{p}, \qquad \mu = t\lambda. \tag{5.1}$$

Then

$$\sum_{A < n \leq A+X} z^{T_{H,y}(n)} \leq X \left(e^{-\mu} + \frac{\mu^{J+1}}{(J+1)!} \right) + (Hy)^J. \tag{5.2}$$

The summation in (5.2) is over integers. In particular, A may be negative and the interval may contain a zero of F_H. No positive-window assumption is used here. The bound includes the empty selected-prime set, $J = 0$, and truncation orders exceeding the number of selected primes.

5.1 Weighted Bonferroni algebra

For a finite index set P and real weights $(w_p)_{p \in P}$, write

$$e_j(w) = \sum_{\substack{S \subseteq P \\ |S| = j}} \prod_{p \in S} w_p, \qquad e_0(w) = 1, \qquad P_J(w) = \sum_{j=0}^{J} (-1)^j e_j(w).$$

Empty products equal one; coefficients of degree greater than $|P|$ are zero. The indices are the elements of P. Equal values of w_p are not deduplicated.

Lemma 5.2 (Three finite inequalities)**.** *For $0 \leq w_p \leq 1$ and even J,*

$$\prod_{p \in P} (1 - w_p) \leq P_J(w) \leq \prod_{p \in P} (1 - w_p) + e_{J+1}(w). \tag{5.3}$$

For nonnegative weights and every natural j,

$$j!\, e_j(w) \leq \left(\sum_{p \in P} w_p \right)^j. \tag{5.4}$$

Finally, if $0 \leq w_p \leq 1$, then

$$\prod_{p \in P} (1 - w_p) \leq \exp\left(- \sum_{p \in P} w_p \right). \tag{5.5}$$

Proof. The weighted Bonferroni inequality (5.3) follows from finite inclusion–exclusion; Appendix A.1 gives the recurrence used in the formalization. This is the usual deterministic weighted inequality, not a statement of independence for the arithmetic events [10]. For (5.4), expand the power on the right as a sum over ordered j-tuples. The tuples with distinct indices contribute exactly $j!e_j(w)$, and all other terms are nonnegative. The case $j = 0$ gives equality. For (5.5), multiply $1 - u \leq e^{-u}$ over the indices, using nonnegativity of all factors. ◻

For the arithmetic weights and their model values, set

$$w_p(n) = t\mathbf{1}_{p \mid F_H(n)}, \qquad b_p = \frac{tH}{p} \quad (p \in \mathcal{P}(H,y)). \tag{5.6}$$

Both sets of weights lie in $[0,1]$, and

$$\prod_p (1 - w_p(n)) = z^{T(n)}, \qquad \sum_p b_p = \mu. \tag{5.7}$$

Applying the first half of (5.3) pointwise gives

$$\sum_{A<n\le A+X} z^{T(n)} \le \sum_{j=0}^{J} (-1)^j \sum_{A<n\le A+X} e_j(w(n)). \tag{5.8}$$

Applying the other half to b, together with (5.4)–(5.5), gives

$$P_J(b) \le e^{-\mu} + \frac{\mu^{J+1}}{(J+1)!}. \tag{5.9}$$

It remains to compare the actual coefficient sums in (5.8) with $XP_J(b)$. The name "model value" does not supply this comparison; the next two steps prove it arithmetically.

5.2 Root counting and arbitrary translated intervals

For a subset $S \subseteq \mathcal{P}(H,y)$, define

$$d(S) = \prod_{p\in S} p, \qquad N_S = \#\{A < n \le A + X : d(S) \mid F_H(n)\}.$$

The integer $d(S)$ is positive and squarefree, including $d(\varnothing) = 1$. For distinct primes, simultaneous divisibility is equivalent to divisibility by the product, also when $F_H(n) = 0$.

Lemma 5.3 (Uniform subset-count discrepancy)**.** *For every such S,*

$$\left| N_S - \frac{XH^{|S|}}{d(S)} \right| \le H^{|S|}. \tag{5.10}$$

Proof. For a prime $p > H$, the congruence $F_H(n) \equiv 0 \pmod p$ holds exactly in the H distinct residue classes $1, \dots, H$. The Chinese remainder theorem therefore gives precisely $H^{|S|}$ root classes modulo $d(S)$.

For every positive integer modulus d and integer representative r, the number of points of that residue class in the translated interval is exactly

$$\#\{A < n \le A + X : d \mid n - r\} = \left\lfloor \frac{A + X - r}{d} \right\rfloor - \left\lfloor \frac{A - r}{d} \right\rfloor. \tag{5.11}$$

Indeed the points have the form $r + jd$, and their allowed integer indices form the half-open interval between the two displayed quotients. Subtracting X/d from (5.11) leaves the difference of two fractional parts, whose absolute value is at most one. Summing the error over the distinct root classes proves (5.10). For $S = \varnothing$, the modulus is one, there is one residue class, and $N_S = X$; the same inequality applies. □

The floor formula is valid for negative A and r, because its divisions are real quotients followed by floors, corresponding to Euclidean integer division in Lean. No truncation-toward-zero convention is being used.

5.3 Summing every retained subset once

The subset interpretation of the elementary coefficients gives

$$\sum_{A<n\le A+X} e_j(w(n)) = t^j \sum_{\substack{S\subseteq \mathcal{P}(H,y)\\ |S|=j}} N_S, \tag{5.12}$$

$$e_j(b) = t^j \sum_{\substack{S\subseteq \mathcal{P}(H,y)\\ |S|=j}} \frac{H^j}{d(S)}. \tag{5.13}$$

Consequently their complete signed discrepancy is

$$\begin{aligned}\mathcal{E}_J &:= \sum_{j=0}^{J}(-1)^j\left(\sum_{A<n\le A+X} e_j(w(n)) - Xe_j(b)\right)\\ &= \sum_{\substack{S\subseteq \mathcal{P}(H,y)\\ |S|\le J}} (-t)^{|S|}\left(N_S - \frac{XH^{|S|}}{d(S)}\right).\end{aligned} \tag{5.14}$$

We take absolute values only after writing this signed identity. Multiplying unrelated one-sided estimates by $(-1)^j$ would not be valid.

Unique factorization makes $S \mapsto d(S)$ injective. Since $y > H \ge 1$, every retained subset satisfies

$$1 \le d(S) \le y^{|S|} \le y^J.$$

Hence the entire retained family has at most $\lfloor y^J \rfloor$ members. From (5.10), $0 < t < 1$, and $H \ge 1$,

$$|\mathcal{E}_J| \le \sum_{\substack{S\subseteq \mathcal{P}(H,y)\\ |S|\le J}} t^{|S|}H^{|S|} \le H^J \lfloor y^J \rfloor \le (Hy)^J. \tag{5.15}$$

This counts each subset once across all degrees. It is the reason the error has no additional factor $J + 1$. For $J = 0$ the only subset is empty, and the bound remains valid.

Completion of Theorem 5.1. Use (5.8), write its right-hand side as $XP_J(b) + \mathcal{E}_J$, and apply (5.9) and (5.15). □

6 Transferring the moment bound to actual candidates

Let $\eta = -\log z > 0$. For a candidate $n > H$, Lemma 4.1 gives $T(n) \le B_H$. If $B_H \ge 0$, monotonicity of z^u for $0 < z < 1$ implies

$$1 \le e^{\eta B_H} z^{T(n)}.$$

Counting at most H candidate integers in the initial window, and applying Theorem 5.1 with $A = 0$, gives the following exact finite transfer.

Theorem 6.1 (Finite candidate-counting inequality)**.** *Under the numerical hypotheses of Theorem 5.1, if $B_H \ge 0$ then*

$$C(X) \le H + e^{\eta B_H}\left\{X\left(e^{-\mu} + \frac{\mu^{J+1}}{(J+1)!}\right) + (Hy)^J\right\}. \tag{6.1}$$

If $B_H < 0$, the separate bound $C(X) \le H$ holds.

The proof uses a necessary condition for every actual candidate, so the count is not limited to an algorithmic survivor set. Also, the multiplier $e^{\eta B_H}$ applies to the *entire* expression inside braces. Discarding it from the factorial tail or the arithmetic remainder would change the theorem. These amplified contributions will be treated separately rather than hidden in an error term.

The two finite main declarations are `Erdos647Sieve.finiteMoment` and `Erdos647Sieve.finiteCounting`. They have no unproved moment, CRT, or distribution assertion as a premise. Their displayed numerical hypotheses, of course, remain part of their types [11].

7 Elementary estimates with the linear term retained

The budget comparison requires more than its leading order. This section keeps the constant contribution coming from the linear term in $\log(H!)$.

7.1 The mandatory small-prime debit

Lemma 7.1 (An elementary reciprocal-prime lower bound)**.** *For every natural $H \geq 2$,*

$$\sum_{p\leq H} \frac{1}{p} \geq \log\log H - 1. \tag{7.1}$$

Here and below an index p denotes a prime.

Proof. Expanding the finite-prime-base Euler product gives

$$E_H := \prod_{p\leq H} (1-1/p)^{-1} = \sum_{\substack{m\geq 1 \\ \text{all prime factors of } m\leq H}} \frac{1}{m} \geq \sum_{m=1}^{H} \frac{1}{m} \geq \log H.$$

The sum over smooth integers converges because only finitely many prime bases are involved and every prime-power geometric series converges. This does not assert convergence of the full harmonic series. Positivity permits the restriction to $1 \leq m \leq H$. For $0 \leq u < 1$, integration of $v/(1-v)$ gives

$$-\log(1-u) - u \leq \frac{u^2}{1-u}.$$

At $u = 1/p$, the right-hand side is $1/[p(p-1)]$. The error over primes is bounded by the telescoping sum

$$\sum_{p\leq H} \frac{1}{p(p-1)} \leq \sum_{m=2}^{H} \left(\frac{1}{m-1} - \frac{1}{m}\right) = 1 - \frac{1}{H} \leq 1.$$

Taking logarithms of the Euler-product comparison and summing the factor bounds yields $\log\log H \leq \log E_H \leq \sum_{p\leq H} 1/p + 1$. □

Corollary 7.2 (Fixed-constant debit)**.** *For every natural $H \geq 2$,*

$$D_H \geq H(\log\log H - 2). \tag{7.2}$$

Proof. Use $\lfloor H/p \rfloor \geq H/p - 1$, sum over the primes at most H, and bound their number by H. Lemma 7.1 then proves the claim. □

The lower bound can be negative for small H; this is harmless. In the formalization the quotients in D_H are natural-number quotients cast to $\mathbb{R}$ only after division. The comparison with real division is a proved rounding lemma.

7.2 Factorials and the uncorrected budget

The proof uses existing Mathlib Stirling machinery, not a new formalization of Stirling's formula [20]. Its convenient consequences are

$$n\log n - n \le \log(n!) \le n\log n - n + \frac{1}{2}\log n + 1 \qquad (n \ge 1), \tag{7.3}$$

and

$$(n/e)^n \le n! \qquad (n \ge 1). \tag{7.4}$$

For instance, the upper estimate follows from monotonicity of the Stirling sequence $n!/[\sqrt{2n}(n/e)^n]$ and its value at $n = 1$; its lower bound $\sqrt{\pi}$ implies the lower estimate in (7.3) after dropping nonnegative logarithmic terms. Only these consequences are needed here.

Put

$$L_{\text{fac}}(H) = \log\left(\frac{(H+2)!}{2}\right).$$

The division is real division after casting the factorial. The exact identity

$$L_{\text{fac}}(H) = \log(H!) + \log(H+1) + \log(H+2) - \log 2 \tag{7.5}$$

and (7.3) give

$$\frac{L_{\text{fac}}(H) - (H\log H - H)}{H} \longrightarrow 0 \qquad (H \in \mathbb{N},\ H \to \infty). \tag{7.6}$$

On the other hand, bounding each floor by its argument and summing logarithms of the positive factors $3, \dots, H+2$ gives

$$Q_H \le \frac{1}{\log 2}\sum_{k=1}^{H}\log(k+2) = \frac{L_{\text{fac}}(H)}{\log 2}. \tag{7.7}$$

Proposition 7.3 (One-sided corrected-budget estimate)**.** *For every $\varepsilon > 0$, eventually over natural H,*

$$\frac{B_H}{H} \le \frac{\log H}{\log 2} - \log\log H + 2 - \frac{1}{\log 2} + \varepsilon. \tag{7.8}$$

Proof. Subtract (7.2) from (7.7), divide by $H > 0$, and use (7.6). □

The term $-1/\log 2$ is the normalized contribution of $-H$ in the factorial estimate. Replacing that estimate by $\log(H!) \le H\log H$ would lose this constant. We do not assert an asymptotic equality for B_H/H: integer floors can contribute a linear total error, and the proof only needs the indicated one-sided bound.

8 The analytic input: convergent Mertens error

Write

$$\vartheta(x) = \sum_{p\le x}\log p, \quad E(x) = \vartheta(x) - x, \quad R(x) = \sum_{p\le x}\frac{1}{p} - \log\log x \qquad (x \ge 2).$$

The classical reciprocal-prime analysis is closely related to the Rosser–Schoenfeld formulas [17]. The formal input used here is supplied by a restricted, audited portion of PNT+ [11, 13]:

$$\exists C \ge 0 \quad \forall x \ge 2, \qquad |E(x)| \le \frac{Cx}{(\log x)^2}. \tag{8.1}$$

The source proves this through its `MediumPNT` theorem. No numerical zero verification or zero-free-region catalogue from `ZetaSummary` is used by the retained route.

Proposition 8.1 (Mertens convergence in the required form)**.** *There is a real number M such that $R(x) \to M$ as $x \to \infty$. Consequently, if u, v are two real-valued cutoffs tending to infinity, then*

$$\sum_{p \le v} \frac{1}{p} - \sum_{p \le u} \frac{1}{p} - \big(\log\log v - \log\log u\big) \longrightarrow 0. \tag{8.2}$$

Proof. For $f(x) = 1/(x \log x)$ on $(1, \infty)$,

$$f'(x) = -\frac{1 + \log x}{x^2 (\log x)^2}.$$

Specializing Abel summation to the prime weights $\log p$ gives the exact identity

$$\sum_{p \le x} \frac{1}{p} = \frac{\vartheta(x)}{x \log x} + \int_2^x \frac{\vartheta(u)(1 + \log u)}{u^2 (\log u)^2}\, du. \tag{8.3}$$

For completeness, the lower endpoint is consistent at $x = 2$: the boundary term is $1/2$ and the integral is zero. There is no omitted endpoint constant. Writing $\vartheta(u) = u + E(u)$ and using the antiderivative

$$G(u) = \frac{1}{\log u} - \log\log u, \qquad G'(u) = u f'(u),$$

rearranges (8.3) to

$$R(x) = G(2) + \frac{E(x)}{x \log x} - \int_2^x E(u) f'(u)\, du. \tag{8.4}$$

By (8.1), the boundary term tends to zero and

$$|E(u) f'(u)| \le C \left(\frac{1}{u (\log u)^3} + \frac{1}{u (\log u)^4} \right).$$

The latter majorant is integrable on $[2, \infty)$. Thus the right-hand side of (8.4) converges, proving existence of M. Composing this convergence with u and v and subtracting proves (8.2). □

The actual value of M is irrelevant: it cancels. A merely bounded error $R(x) = O(1)$ would leave an uncontrolled constant in (8.2) and would not suffice for the later positive gap.

For real $y \ge H$, the exact selected-prime identity is

$$\sum_{p \in \mathcal{P}(H, y)} \frac{1}{p} = \sum_{p \le y} \frac{1}{p} - \sum_{p \le H} \frac{1}{p}. \tag{8.5}$$

The formalization proves this finite-set partition and then the corresponding limit for the actual selected-prime definition. The PNT+ constants and interfaces are hidden behind the analytic compatibility boundary; downstream parameter proofs consume (8.2), not upstream internals.

9 Parameter asymptotics and the positive mass–budget gap

For the remainder of the proof X tends to infinity through natural numbers. Use the fixed choices

$$\begin{gathered} Q = \log X, \qquad L = \log Q, \qquad H = \lfloor Q^a \rfloor, \qquad J = 2 \left\lceil \frac{H}{100} \right\rceil, \\ y = X^{1/(4J)}, \qquad t = \frac{1}{1000 L}, \qquad z = 1 - t. \end{gathered} \tag{9.1}$$

These are the existing formal parameter definitions. In particular, the natural expression for J is $2((H + 99)/100)$, with natural division. Its equality to the ceiling expression in (9.1) is proved separately.

9.1 Why this exponent is selected

If a provisional window were $H \asymp Q^b$ and $J \asymp H$, then $\log y \asymp Q^{1-b}$. The leading prime mass per shift would have coefficient $1-b$ in front of L, while the logarithmic budget would have coefficient $b/\log 2$. The equality

$$\frac{a}{\log 2} = 1 - a \tag{9.2}$$

selects $a = \log 2/(1+\log 2)$. At this boundary the leading logarithms cancel, so the secondary terms and their fixed constants matter. This identifies the critical balance of the chosen argument, not an optimality theorem among all possible methods.

Lemma 9.1 (Actual window, truncation, and cutoff limits)**.** *For the parameters in* (9.1)*,*

$$H \to \infty, \qquad H/Q^a \to 1, \qquad \log H - aL \to 0, \tag{9.3}$$

$$H/50 \le J < H/50 + 2, \qquad J/H \to 1/50, \tag{9.4}$$

$$\frac{\log y}{Q^{1-a}} \to \frac{25}{2}, \qquad \log\log y - (1-a)L \to \log(25/2). \tag{9.5}$$

Moreover $y \to \infty$, $H < y$ eventually, J is even, and all numerical hypotheses of Theorem 5.1 hold eventually.

Proof. The inequalities $Q^a - 1 < H \le Q^a$ prove the first two assertions in (9.3). For sufficiently large X the quantities are positive, and $\log H - aL = \log(H/Q^a) \to 0$. The exact ceiling identity for J proves (9.4).

For positive X and J, the real-power definition gives $\log y = Q/(4J)$. Hence

$$\frac{\log y}{Q^{1-a}} = \frac{Q^a}{4J} \longrightarrow \frac{25}{2}.$$

Taking logarithms proves the second assertion in (9.5). The power Q^{1-a} tends to infinity and dominates L. Therefore $\log y \to \infty$ and $\log H/\log y \to 0$, so $y \to \infty$ and $H < y$ eventually. Finally, $L \to \infty$ implies $0 < t \le 1/2$ eventually, and thus $0 < z < 1$. Positivity of H, J and evenness of J have already been established. □

These arguments use familiar limits, but their formal application requires the floor, ceiling, cast, and nonzero-denominator obligations. They are supplied for these specific functions of X, rather than left as cutoff hypotheses.

Proposition 9.2 (Prime-mass expansion)**.** *Let $\lambda = \lambda(H, y)$ for* (9.1)*. Then*

$$\frac{\lambda}{H} = (1-a)L - \log\log H + \log(25/2) + o(1). \tag{9.6}$$

Proof. The two cutoffs H, y tend to infinity and satisfy $H \le y$ eventually. Apply (8.2) and (8.5) to obtain $\lambda/H = \log\log y - \log\log H + o(1)$, and then use (9.5). □

Proposition 9.3 (Fixed positive gap and size bounds)**.** *Define*

$$K = \log(25/2) + \frac{1}{\log 2} - 2. \tag{9.7}$$

For every fixed real $\delta < K$, eventually $\lambda - B_H \ge \delta H$. In particular,

$$\lambda - B_H \ge \frac{3}{2}H, \qquad 0 \le \lambda \le HL, \qquad B_H \le HL \quad \textit{eventually}. \tag{9.8}$$

Also $\lambda/(HL) \to 1-a$.

Proof. Let
$$F(H) = \frac{\log H}{\log 2} - \log\log H + 2 - \frac{1}{\log 2}.$$
From (9.6), (9.3), and (9.2),
$$\frac{\lambda}{H} - F(H) \longrightarrow K. \tag{9.9}$$
By Proposition 7.3, for every $\varepsilon > 0$ one has $B_H/H \le F(H) + \varepsilon$ eventually, also after composing with $H(X)$. Given $\delta < K$, choose $\varepsilon = (K - \delta)/3$. Eventually (9.9) is at least $K - \varepsilon$, so $(\lambda - B_H)/H \ge K - 2\varepsilon > \delta$. The certified elementary logarithm estimate $K > 3/2$ is recorded in Appendix A.2.

For the size bound, $\log H/L \to a$, and
$$\frac{\log\log H}{L} = \frac{\log\log H}{\log H}\,\frac{\log H}{L} \longrightarrow 0.$$
Dividing (9.6) by L gives $\lambda/(HL) \to 1 - a < 1$. The mass is nonnegative by definition; hence $0 \le \lambda \le HL$ eventually. The positive gap gives $B_H \le \lambda$, which yields the remaining upper bound. □

There is no assertion that $(\lambda - B_H)/H$ converges to K: only (9.9) has been shown to converge, and the budget estimate is one-sided. Nor is any sign hypothesis on B_H part of Proposition 9.3.

10 Every amplified contribution

We now bound the four terms in Theorem 6.1, using (9.1). Write
$$\eta = -\log(1-t), \qquad \mu = t\lambda, \qquad A_X = \eta B_H.$$
The symbol A_X in this section is an amplification exponent, not the interval translation used earlier. For clarity, all inequalities involving the positive multiplier are first proved in the case $B_H \ge 0$.

10.1 Principal saving and the source of 1499

For $0 \le t \le 1/2$, elementary calculus gives
$$0 \le -\log(1-t) \le t + t^2 \le 2t. \tag{10.1}$$
For example the derivative of $t + t^2 + \log(1-t)$ is $t(1-2t)/(1-t) \ge 0$ on this interval, and the function vanishes at zero. Multiplying (10.1) by the nonnegative budget and using (9.8) gives
$$A_X \le 2tHL = H/500, \qquad 0 \le \mu \le tHL = H/1000. \tag{10.2}$$
More importantly,
$$\begin{aligned} A_X - \mu &\le (t+t^2)B_H - t\lambda \\ &= -t(\lambda - B_H) + t^2 B_H \\ &\le -\frac{3}{2}tH + t^2 HL = -\frac{1499}{10^6}\frac{H}{L}. \end{aligned} \tag{10.3}$$
Thus the principal contribution is at most $Xe^{-\kappa H/L}$. The integer 1499 is simply $1500 - 1$: the first part is the gap saving and the second pays for the quadratic logarithm correction. It is not an invariant of problem #647 and is not claimed to be optimal.

10.2 The factorial tail with a growing numerator

Put $q = J + 1$. By (9.4), $q \geq H/50$ and $q \geq 1$ eventually. Equation (10.2) gives $\mu/q \leq 1/20$. Consequently (7.4) yields

$$\frac{\mu^q}{q!} \leq \left(\frac{e\mu}{q}\right)^q \leq (e/20)^q. \tag{10.4}$$

Since $\log 20 - 1 > 0$ and $q \geq H/50$,

$$\begin{aligned} e^{A_X}\frac{\mu^q}{q!} &\leq \exp\left(\frac{H}{500} - (\log 20 - 1)q\right) \\ &\leq \exp\left[-H\left(\frac{\log 20 - 1}{50} - \frac{1}{500}\right)\right] \leq e^{-H/30}. \end{aligned} \tag{10.5}$$

The last rational comparison is included in Appendix A.2. This is a bound on the actual growing numerator $\mu(X)$; the elementary limit $u^q/q! \longrightarrow 0$ for a *fixed* u would not justify this step.

10.3 Arithmetic remainder and the exceptional window

For the actual positive parameters, the power laws give the exact identity

$$y^J = X^{1/4}. \tag{10.6}$$

Keeping the full multiplier on the discrepancy term,

$$e^{A_X}(Hy)^J = \exp\left(\frac{Q}{4} + J\log H + A_X\right). \tag{10.7}$$

Since $J \asymp H \asymp Q^a$ and $a < 1$,

$$\frac{J\log H + H/500}{Q} \longrightarrow 0.$$

It follows from (10.2)–(10.7) that

$$e^{A_X}(Hy)^J \leq e^{Q/2} = \sqrt{X} \qquad \text{eventually.} \tag{10.8}$$

Independently $H \asymp Q^a \leq \sqrt{X}$ eventually, so the exceptional initial window also contributes at most $\sqrt{X}$.

Theorem 10.1 (All-sign intermediate bound)**.** *For all sufficiently large natural X,*

$$\boxed{C(X) \leq Xe^{-\kappa H/L} + Xe^{-H/30} + 2\sqrt{X}.} \tag{10.9}$$

There is no assumption on the sign of B_H in this statement.

Proof. For $B_H \geq 0$, apply Theorem 6.1 and the three estimates (10.3), (10.5), and (10.8), along with the exceptional-window bound. For $B_H < 0$, (4.6) gives $C(X) \leq H \leq \sqrt{X}$, which is smaller than the displayed nonnegative right-hand side. The finitely many eventual conditions used in these comparisons can be imposed simultaneously. □

The formal declaration is `Erdos647Sieve.Endpoint.eventually_candidateCount_le_amplified_errors`. It is the last counting input to the final assembly. Every error source from the finite theorem has a named bound before that assembly is attempted.

11 Final absorption and the strict coefficient range

Let $U = Q^a$ and $S = U/L$. The parameter limits give

$$S \to \infty, \quad H/U \to 1, \quad S/H \to 0, \quad S/Q \to 0. \tag{11.1}$$

For example $S/H = (U/H)/L \to 0$, and $S/Q = Q^{a-1}/L \to 0$. All denominators are positive eventually.

Proof of Theorem 2.1. Fix $0 < c < \kappa$. Divide the right-hand side of (10.9) by the positive quantity Xe^{-cS}. The exact resulting ratio is

$$\exp\left[\left(c - \kappa\frac{H}{U}\right)S\right] + \exp\left[\left(c\frac{S}{H} - \frac{1}{30}\right)H\right] + 2\exp\left[\left(c\frac{S}{Q} - \frac{1}{2}\right)Q\right]. \tag{11.2}$$

By (11.1), the three coefficients in parentheses tend respectively to $c - \kappa < 0$, $-1/30$, and $-1/2$. Each is eventually bounded above by a fixed negative number. Their multiplying scales S, H, Q tend to infinity. Therefore every term of (11.2) tends to zero, and the sum is eventually at most one. Combining this with Theorem 10.1 proves (2.2). Finally $0 < 1/1000 < 1499/10^6$ is rational arithmetic, so the stated specialization requires no additional analytic estimate. □

This proof absorbs the rounding of H, the factor two, and both secondary terms without an unspecified multiplicative constant in the conclusion. Its limit statements assert the existence of an onset; they do not compute that onset.

11.1 Why the boundary coefficient is not asserted

At $c = \kappa$, the first ratio in (11.2) becomes

$$\exp\left(\kappa\frac{U - \lfloor U\rfloor}{L}\right) \geq 1. \tag{11.3}$$

The other two terms are positive. Thus the available majorant cannot itself be bounded by $Xe^{-\kappa S}$ using this argument. This does not prove that the actual candidate count violates such a bound; it shows why the established coefficient range is strict. Taking c to the boundary would also require control of the dependence of $X_0(c)$ on c.

The right presentation is therefore the complete range $0 < c < \kappa$, followed by a simple explicit corollary. The parameter $t = 1/(1000L)$ and the final coefficient c have different roles. The former tunes the moment; the latter is the saving that survives all losses. Here the original $c = 1/1000$ target happens to follow for free, but the proof never needs to optimize that arbitrary value.

12 Reusable mathematical and formal machinery

The proof yields more reusable material than the final statement alone. This section separates existing library components from natural extensions that have not been exported as Lean theorems in the audited snapshot.

12.1 Components already implemented

The finite package provides weighted Bonferroni inequalities for finite lists, coefficient bounds, and a bridge from list coefficients to subsets of a finite index set. These do not depend on the divisor problem.

The distinction between indices and weight values allows repeated weights without accidental loss of multiplicity. They are suitable building blocks for deterministic lower-tail estimates in other finite sieves.

The progression-counting result (5.11) is uniform in the translation and residue representative. The prime-subset product lemmas provide injective encoding by positive squarefree products. The combination of these with CRT supplies exact local densities and an explicit cumulative error. Their useful abstraction is not "primes behave independently", but "simultaneous congruences have exact finite root counts, and interval sampling has a controlled error".

The seven elementary modules promoted to the Mathlib-only package expose factorial bounds and limits, the finite Euler-product comparison, the reciprocal-prime lower bound, the mandatory debit, and the signed-budget estimate. Their public facade is `Erdos647Sieve.Elementary`. The finite theorem is accessible through `Erdos647Sieve.Finite`. Neither facade requires PNT+ [11].

The analytic package supplies Mertens convergence and its two-cutoff cancellation in the precise form consumed by the application. Its public facades are `Erdos647Research.Analytic` and `Erdos647Research.Endpoint`. The latter exposes the completed endpoint; "Research" in the module name indicates its separate dependency boundary, not an unfinished proof. Only the compatibility layer knows the upstream PNT+ interfaces.

12.2 A general residue-set template

The following is a mathematical extension of the finite argument, stated to make the transfer mechanism explicit. It is *not* claimed as a separately exported generic theorem in the v48 Lean package. Let P be any finite set of distinct primes. At each $p \in P$ choose a residue set $R_p \subseteq \mathbb{Z}/p\mathbb{Z}$ of size ρ_p. Define

$$T_R(n) = \sum_{p \in P} \mathbf{1}_{n \bmod p \in R_p}, \qquad \rho(S) = \prod_{p \in S} \rho_p, \qquad d(S) = \prod_{p \in S} p.$$

For $0 < z < 1$, $t = 1 - z$, even $J \geq 0$, and an interval of X consecutive integers, the same proof gives

$$\sum_{A < n \leq A+X} z^{T_R(n)} \leq X \left(e^{-\mu_R} + \frac{\mu_R^{J+1}}{(J+1)!} \right) + \sum_{\substack{S \subseteq P \\ |S| \leq J}} t^{|S|} \rho(S), \qquad \mu_R = t \sum_{p \in P} \frac{\rho_p}{p}. \tag{12.1}$$

Indeed CRT gives $\rho(S)$ simultaneous classes modulo $d(S)$, the counting error is at most $\rho(S)$, and the model weights $t\rho_p/p$ lie in $[0, 1]$. Equations (5.3)–(5.14) then give (12.1). This proof on paper specifies the additional generic interface one could formalize; it does not silently generalize existing theorem types. Bounding the last sum still requires information on the particular residue sets and cutoff products.

12.3 Other budget profiles and practical limits

For restrictions $\tau(n - k) \leq f(k)$ with $f(k) \geq 1$, the same elementary argument suggests the budget

$$\sum_{k=1}^{H} \left\lfloor \frac{\log f(k)}{\log 2} \right\rfloor - D_H.$$

For a nonconsecutive family of shifts, both the mandatory small-prime debit and the local root count must be recalculated. If several shifts coincide modulo p, the root density is no longer H/p. If prime powers are included as sieve moduli, prime-subset product injectivity alone is no longer the correct bookkeeping device. These are genuine extension obligations.

The final ratio-absorption pattern is another broadly useful component: prove a normalized error tends to zero, retain a fixed gap in the exponent, and only then absorb finitely many positive contributions. It avoids a fragile sequence of unnecessarily sharp numerical thresholds. Our implementation keeps the relevant scale comparisons and the final assembly in separate modules.

13 Other Erdős problems: concrete connections and limitations

The links below are proposed uses of the machinery, not new results on the listed problems. The statements are taken from the original problem literature and the modern problem catalogue, with contemporary comparisons to Tao–Teräväinen and Lau. A common arithmetic vocabulary does not make an existence theorem a consequence of an upper-counting bound.

13.1 Problem #679: smaller backward prime-factor barriers

Problem #679 asks, for each $\varepsilon > 0$, whether infinitely many n satisfy

$$\omega(n-k) < (1+\varepsilon)\frac{\log k}{\log\log k}$$

for all $k < n$ sufficiently large depending only on ε [4, 7]. The quantifier on the initial cutoff is important: it must be independent of n.

This is a close structural match to the backward-window bookkeeping. A possible first application is a finite necessary-condition count over shifts $K_\varepsilon \le k \le H$, with the aggregate barrier replacing Q_H and the mandatory debit recomputed for the shorter shifted window. The weighted coefficient and CRT machinery would then be available. Such an application could quantify the scarcity of integers satisfying a prescribed finite set of barriers. It would not, by itself, settle whether infinitely many full barrier solutions exist, nor may one let K_ε grow with n without changing the question. Lau discusses sharper barriers in connection with a conditional Cramér-type model; those conditional conclusions are not inputs to our proof [14].

13.2 Problem #826: linear forward divisor barriers

Problem #826 asks whether infinitely many n have $\tau(n+k) \ll k$ for every $k \ge 1$, with a uniform implied constant [5]. Lau obtains a polynomial barrier $\tau(n+k) \ll k^C$ for an absolute C, not the requested linear one [14].

A forward window admits the same local congruence counting after reversing the shift convention. For a fixed proposed divisor-barrier constant, the logarithmic-budget conversion could yield finite upper bounds for its survivors, with constants and the length of the window kept explicit. The limitation is decisive: $2^{\omega(m)} \le \tau(m)$ gives a necessary condition for small τ, not a sufficient one. Constructing many integers with small divisor counts also requires control of prime multiplicities and cofactors. The current sieve does not provide that construction.

13.3 Problem #413: the barrier $\omega(n-k) \le k$

Problem #413 concerns infinitely many integers obeying backward prime-factor barriers of the form $\omega(n-k) \le k$ for $1 \le k < n$ [2, 18]. The same finite incidence-counting and congruence modules could formalize a relaxation of these constraints. But summing this barrier gives a budget of order H^2, rather than the $H \log H$ budget used in this paper. Thus neither the exponent a nor the mass–budget gap transfers automatically. A useful project here would be a formal, parameter-dependent finite counting theorem for that exact barrier, followed by an independent assessment of whether it says anything substantive about the infinite existence question.

13.4 Problem #248: a solved benchmark for reusable formalization

Problem #248 asks for infinitely many n with a uniform linear forward bound on the number of prime factors. Tao–Teräväinen prove $\omega(n+k) \le \Omega(n+k) \ll k$ for every positive k [1, 18]; Lau subsequently obtains a logarithmic barrier for $k \ge 2$ [14].

Here the prospective benefit is formalization, not another claimed solution. The finite weighted algebra, local root counting, and exponential-error interfaces could serve as independently usable components of a formalized proof or companion quantitative estimate. The existence arguments in those papers need additional probabilistic and sieve ingredients. Our deterministic upper-moment estimate cannot replace those ingredients merely because both arguments discuss prime factors of nearby integers.

13.5 A disciplined transfer criterion

For each proposed application, three tasks should precede a large new formalization: identify the exact arithmetic barrier and its quantifiers; compute the local root densities and compulsory debit; and determine whether the desired conclusion is an upper bound, an existence result, or a uniform all-shifts assertion. Only then is reuse more than a similarity of notation. The present work is strongest as a library for finite counting and audited analytic assembly. It does not furnish a general mechanism for resolving all prime-factor problems.

14 Conclusion

The divisor restrictions in Erdős #647 impose a logarithmic bound on the total number of distinct prime-factor occurrences in a growing backward window. Removing the unavoidable small-prime occurrences leaves a signed budget that can be compared with a precisely evaluated selected-prime mass. A finite weighted Bonferroni argument, exact CRT root counts, and an injective encoding of all retained prime subsets control the averaging error without an extra factor depending on the truncation order.

At the exponent $a = \log 2/(1 + \log 2)$, the leading terms cancel. Mertens error cancellation and the retained linear factorial term leave a fixed positive mass–budget gap. The complete amplified principal term, factorial tail, arithmetic remainder, and exceptional window then give Theorem 10.1. Final ratio absorption proves Theorem 2.1 for every fixed $0 < c < 1499/10^6$, including $1/1000$.

The audited formalization and the human-readable proof have complementary roles. Lean fixes exact hypotheses, signs, boundary cases, and dependencies; the exposition identifies the mathematical mechanism and its limitations. The AI–operator workflow produced the development, but its claims rest on the statements and proof artifacts rather than on the model's authority. The reusable finite and elementary library is separate from the analytic package, and the latter imports only the audited PNT+ slice it needs.

What remains is not a missing step in this endpoint deduction, but a different question: can one exclude every candidate beyond 24, or construct one? The bound proved here still allows an infinite sparse set. External comparison of novelty, mathematical review, broader independent proof replay, and any extension to related problems should therefore be pursued as distinct tasks, not conflated with a resolution of #647.

Acknowledgments and contributions

The project uses the work of the Lean, Mathlib, and PNT+ communities, and the public problem curation of Thomas F. Bloom. Existing mathematical and formal sources are credited in the bibliography and the repository. GPT 6 Astra is the AI author attribution requested for this paper. Justin V. Wick operated the interactive workflow, executed the verification runs, supplied the returned artifacts, and managed the project. No institutional affiliation or endorsement by the cited communities is implied.

A Technical details and boundary cases

A.1 The finite Bonferroni recurrence

This supplies a direct algebraic proof of (5.3). For a finite list w, write $V(w) = \prod_i (1 - w_i)$ and $E_J(w) = P_J(w) - V(w)$. Inserting a new weight u gives

$$e_{j+1}(u :: w) = e_{j+1}(w) + ue_j(w), \qquad P_{j+1}(u :: w) = P_{j+1}(w) - uP_j(w).$$

It follows that

$$E_{j+1}(u :: w) = E_{j+1}(w) - uE_j(w).$$

The invariant $(-1)^j E_j(w) \geq 0$ is proved by induction on the list. At $j = 0$, $E_0 = 1 - V \geq 0$ because $0 \leq V \leq 1$. At $j + 1$, multiplying the recurrence by $(-1)^{j+1}$ expresses it as the sum of the two nonnegative quantities $(-1)^{j+1}E_{j+1}(w)$ and $u(-1)^j E_j(w)$. For even J this proves $P_J \geq V$. The next truncation satisfies $P_{J+1} = P_J - e_{J+1} \leq V$, giving the upper half of (5.3). This also covers the empty list and orders above its length.

A.2 The elementary logarithm constants

The numerical arguments need only coarse certified bounds

$$0.693 < \log 2 < 0.7.$$

These follow from the explicit real-logarithm bounds in the pinned Mathlib sources [20]. Since $25/2 > 8$,

$$K > 3\log 2 + \frac{1}{\log 2} - 2 > 3\frac{693}{1000} + \frac{10}{7} - 2 = \frac{10553}{7000} > \frac{3}{2}.$$

Similarly $20 > 16$ implies $\log 20 > 4\log 2 > 2772/1000$, whence

$$\frac{\log 20 - 1}{50} - \frac{1}{500} > \frac{1.772}{50} - \frac{1}{500} = \frac{209}{6250} > \frac{1}{30}.$$

All decimals in these displayed comparisons denote exact rational numbers. No floating-point numerical evaluation is a proof dependency. Sharpening these inequalities is unnecessary for the stated endpoint.

A.3 Boundary cases recorded in the formal interfaces

Case	Treatment
Negative interval translation	The finite moment uses integer intervals and Euclidean quotient counts; no positivity is imposed on the translation.
Zeros of $F_H(n)$	Divisibility and CRT root identities apply to all integer inputs, including zero values of the polynomial.
Empty prime set or subset	Empty products equal one; the empty subset has modulus one and exact count X.
$J = 0$ or $J > \|\mathcal{P}\|$	The finite Bonferroni and coefficient conventions cover these cases. Endpoint parameters eventually have $J > 0$.
Negative corrected budget	A separate exclusion gives $C(X) \le H$; the positive-weight estimates are not multiplied by a negative budget.
Coefficient at the boundary	Only $c < \kappa$ is asserted. Equation (11.3) shows the limitation of the current majorant.
A translated endpoint	The finite moment is uniform in translation. No separate all-translations endpoint theorem is claimed as an exported result here.

B Formalization map and verification provenance

B.1 Theorem-to-source correspondence

The following map identifies representative exported or audited declarations; it is not a claim that every sentence of the exposition is a separate Lean lemma. In the table, the prefixes are S for `Erdos647Sieve`, E for `Erdos647Sieve.Elementary`, A for `Erdos647Sieve.Analytic`, and P for `Erdos647Sieve.Endpoint`. The complete source snapshot is in the accompanying verification artifact [11].

Mathematical component	Representative Lean declaration
Candidate and finite claims	`S.Candidate`; `S.FiniteMomentClaim`; `S.FiniteCountingClaim`
Mandatory debit, Section 4	`S.budgetDebit`
Weighted inclusion–exclusion, Section 5	`S.Bonferroni.even_sum_bounds`
Finite moment and transfer, Sections 5–6	`S.finiteMoment`; `S.finiteCounting`
Reciprocal-prime lower bound	`E.prime_reciprocal_lower`
Fixed-constant debit	`E.smallPrimeDebit_lower`
Factorial lower bound and shifted limit	`E.factorial_pow_lower`; `E.shiftedFactorialLog_residual_tendsto_zero`
Budget estimate, Proposition 7.3	`E.correctedBudget_upper_eventually`
Mertens convergence	`A.reciprocalPrimeLimit_exists`
Two-cutoff error cancellation	`A.selectedPrimeSum_difference_tendsto_zero`
Actual prime-mass expansion	`P.endpoint_primeMass_expansion_tendsto_zero`
Positive mass–budget gap	`P.eventually_primeMass_sub_correctedBudget_ge`
Mass and budget sizes	`P.eventually_endpoint_mass_budget_bounds`
Principal exponent	`P.eventually_amplification_exponents_le`
Amplified factorial tail	`P.eventually_amplified_moment_tail_le`
All-sign counting bound	`P.eventually_candidateCount_le_amplified_errors`
Final absorption	`P.eventually_amplifiedErrorMajorant_le_endpointRHSWith`

Mathematical component	Representative Lean declaration
General coefficient theorem	`P.endpointBound_of_lt_principal_rate`
Explicit and historical corollaries	`P.endpointBound_one_div_thousand`; `P.positiveEndpoint`; `S.endpoint`

The final assembly is intentionally short. In outline, it combines the all-sign majorant with its absorption into the desired endpoint expression:

```
theorem endpointBound_of_lt_principal_rate
  (c : Real) (hc : c < (1499 / 1000000 : Real)) :
  EndpointBound c
```

The printed type is not the proof; the supplied source contains the proof body, and the final audit checks both the expanded endpoint definition and the transitive axioms. In particular c is not an assumed bound on a finite majorant: that bound has already been proved in the imported absorption module.

B.2 Exact snapshot and trust record

The development URL is `https://github.com/JustinWick/erdos647`. The paper's precise basis is the operator-supplied v48 archive identified below, rather than an assertion that a mutable repository branch always contains the same bytes. The recorded UTC run spans `2026-09-09 01:40:11–01:45:24`.

Recorded item	Value
Lean toolchain	`leanprover/lean4:v4.32.2`
Platform in compiler report	`x86_64-unknown-linux-gnu`
Configured proof gates	30 passed; none failed or blocked
Distinct audited declarations	235
Exact-type/definition examples	220 in 32 audit files
Engineering tests	317 passed
Executed commands	70, all with zero exit status
Build and audit diagnostics	No warnings or errors found in the reviewed logs
Archive manifest	296 entries, all matching their recorded hashes
Source snapshots	179 checked inputs before and after; unchanged
Retained PNT+ source closure	14 modules; admission scan passed
Axiom policy	Exactly the allowed set in (3.1); no unauthorized axiom reported

Version identifiers.

Lean compiler commit:

`f3b06c705e6c85f5314019d5d3baab0fec5b580c`

Mathlib revision:

`905b95818eb32af7874a58b427f50c1711a5e96c`

PNT+ revision:

`a5154676af9aa3095150ee410cdda80555aa0642`

Artifact and source hashes.

Run archive:
`erdos647_repo_v48_results_20260909T014011Z_dd165b9f.zip`
SHA-256:
`7043421f1ec2ca5c85963e84a1a670096b23a2688fb16932a42b738c1343561e`

`Erdos647Sieve/Specification.lean`:
`39d7c9e69faba89c27d738c6bf37f8b363197e2e5942281c5aedbbba7374c574`

`research/Erdos647Research/Endpoint/Main.lean`:
`91700a3fe721917602595cbd80ba1a515ada11abc679106b37f4167426dec0b4`

The review that produced these counts verifies the supplied archive and its logs; it is not another Lean execution. The final endpoint had already passed in the v47 proof checkpoint. The v48 record additionally checks the promoted Mathlib-only elementary modules and the public analytic and endpoint facades. Thus the packaging changes are not confused with a new mathematical theorem.

B.3 Reproducibility and the public interface

The repository contains pinned toolchain and Lake manifests, ordinary core build/test targets, and separate research targets. The top-level runner manages incremental checking, setup when explicitly requested, and diagnostic archives. A mathematical consumer can instead import the public facades without knowing the historical overlay sequence. The core and analytic packages maintain separate dependency boundaries; only the latter needs PNT+.

The companion paper-source bundle includes this LaTeX source, bibliography, a PDF build script, the evidence-review report, and the original v48 diagnostic archive. It is sufficient to identify the proof snapshot described here. Public deployment, independent kernel replay, and peer review are not inferred from the presence of a CI configuration or from file hashes.

C Notation and proof dependencies

Symbol	Meaning
τ, ω, Ω	Divisor count; number of distinct prime factors; number with multiplicity
$C(X)$	Number of actual full-condition candidates $24 < n \le X$
$F_H(n)$	$\prod_{k=1}^{H}(n-k)$, the window polynomial
$\mathcal{P}(H, y)$	Primes $H < p \le y$
$T_{H,y}(n)$	Number of selected primes dividing $F_H(n)$
Q_H, D_H, B_H	Logarithmic budget, mandatory small-prime debit, signed difference
λ	Prime mass $H\sum_{H<p\le y} 1/p$
Q, L, U, S	$\log X$, $\log\log X$, Q^a, U/L
H, J, y	$\lfloor Q^a \rfloor$, $2\lceil H/100 \rceil$, $X^{1/(4J)}$
t, z, η, μ	$1/(1000L)$, $1-t$, $-\log z$, $t\lambda$
K	$\log(25/2) + 1/\log 2 - 2$, the available gap constant
κ	$1499/10^6$, the upper threshold of the proved open coefficient range
c	A fixed positive coefficient strictly below κ

The logical dependencies may be read as two converging chains. The finite chain is: weighted Bonferroni algebra and CRT progression counts imply the finite moment; that moment together with the signed divisor budget implies the finite counting inequality. The analytic chain is: the PNT/integrability input implies Mertens convergence; actual cutoff asymptotics and the elementary factorial/debit estimates imply the mass–budget gap. The gap controls the amplification of the finite inequality, and the scale limits absorb all remaining contributions. No arrow in these chains stands for an unproved premise in the final Lean theorem.

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